# Ch. 5 CW Pt. 1

Principles of Microeconomics

September 21, 2025

## Instructions

Please answer on a separate sheet of paper and submit on Gradescope by the end of class.

1. Do you think the price elasticity of demand for Ford SUVs will increase, decrease, or remain the same when each of the following events occurs? Please explain your answer.

   (a) Other car manufacturers, such as General Motors, decide to make and sell SUVs.

   (b) SUVs produced in foreign countries are banned from the U.S. market.

   (c) Due to ad campaigns, Americans believe that SUVs are much safer than ordinary passenger cars.

   (d) The time period over which you measure the elasticity lengthens. During that longer time, new models such as four-wheel-drive cargo vans appear.

2. Suppose demand for gas in September is given by the following demand schedule:

| Price ($ per gallon) | Quantity Demanded (millions of gallons) |
|---|---|
| 5 | 5 |
| 4 | 6 |
| 3 | 7 |
| 2 | 8 |
| 1 | 9 |

   (a) Calculate the price elasticity of demand between a price of $5 and $4 per gallon.

   (b) Is demand for gas elastic or inelastic?

   (c) If the price of gas changed in September, but we measured the change in quantity demanded in March of the next year, would demand for gas be more or less elastic? Why?

3. (a) Suppose the price elasticity of supply for iPhones is 1.6. If the price of iPhones decreases by 18; what happens to the quantity supplied?

   (b) Suppose the income elasticity of demand for iPhones is 0.6. If average income increases from $38,000 to $42,000, what happens to the quantity of iPhones demanded?

   (c) What sign would the cross-price elasticity of demand between iPhones and portable chargers have?

   (d) Suppose Apple raises the price of an iPhone by 5% and iPhone revenue increases by 2%. What is the price elasticity of demand.

4. Taiwan is a major world supplier of semiconductor chips. A recent earthquake severely damaged the production facilities of Taiwanese chip-producing companies, sharply reducing the amount of chips they could produce. Assume that the total revenue of a typical non-Taiwanese chip manufacturer rises due to these events. In terms of an elasticity, what must be true for this to happen? Illustrate the change in total revenue with a diagram?